Automated Credit Recovery Architecture (.NET Core + React)

1. Recommended Design Patterns

- CQRS (Command Query Responsibility Segregation):

  Separate read and write operations to improve scalability and performance.

- Event Sourcing:

  Maintain immutable logs of all recovery actions for audit and compliance.

- Saga Pattern:

  Coordinate long-running transactions across microservices (e.g., payment reconciliation, legal escalation).

- API Gateway Pattern:

  Central entry point for client requests, handling routing, authentication, and throttling.

- Observer Pattern:

  Notify dependent services (e.g., analytics, compliance) when recovery events occur.

2. Microservices Layering

- Presentation Layer (React):

  User dashboards, agent portals, and customer self-service interfaces.

- API Gateway Layer (.NET Core):

  Handles authentication, routing, and request aggregation.

- Service Layer (.NET Core Microservices):
  - Communication Service: SMS/WhatsApp/Voice integration.
  - Recovery Workflow Service: Automation rules, escalation logic.
  - Payment Reconciliation Service: Sync with core banking APIs.
  - Legal Service: Auto-preparation of legal documentation.
  - Analytics Service: Portfolio health, agent performance, automation efficacy.

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- Data Layer:

  SQL/NoSQL databases, event store for audit logs, integrated with CRM and core banking.

3. Communication Flow

- React frontend communicates with API Gateway via REST/GraphQL.
- API Gateway routes requests to appropriate microservices.
- Microservices communicate asynchronously via message broker (e.g., RabbitMQ, Kafka).
- Event sourcing ensures all actions are logged and replayable.
- Analytics service consumes events for real-time dashboards (Power BI integration).
- Compliance service subscribes to events for NDPR/CBN/Shariah checks.

4. Implementation Roadmap

- Phase 1: Architecture setup (.NET Core microservices, React frontend, API Gateway).
- Phase 2: Develop core services (Recovery Workflow, Communication, Payment Reconciliation).
- Phase 3: Integrate analytics and compliance services.
- Phase 4: Pilot rollout with limited portfolio.
- Phase 5: Full rollout with optimization and scaling.